

Halo Size

Consider the scattering at some instant of time after the arrival of the unscattered radiation. The surface of equal time delay is an ellipsoid of revolution (around the axis through us and the source; source and observer occupy the two foci). If the source is infinitely far away, that ellipsoid degenerates into a paraboloid (with us at the focus). At any given instant, we will see X-rays arrive that have been scattered by dust along that paraboloid. At later times, we see X-rays scattered by dust along a bigger paraboloid.

Assuming a single scattering site, at distance d (in practice, as shown below, this will be at the most distant dust that significantly contributes). The light travel time delay Δt is related to the distance d to the scattering dust and the opening angle (radius) θ of the halo by (for small θ)

$$\Delta t = \frac{1}{2} \left(\frac{d}{c} \right) \theta^2 \quad (1)$$

Invert the relation for θ :

$$\theta = \left(\frac{2c\Delta t}{d} \right)^{1/2} = 0.49 \left(d/8500 \text{ pc} \right)^{-1/2} (\Delta t/1 \text{ yr})^{1/2} \text{ degrees} \quad (2)$$

For scattering by dust close to the Galactic Plane, the distance to the most distant contributing dust should be of order, to a few times, the distance to the Galactic Center, 8500 parsec. For time delays of order a year, halos should have a radius of order half a degree.

The scaling with time is completely straightforward—we know exactly how long ago a given GW event occurred, for any X-ray dataset we are looking at. The scaling with distance is a bit trickier. In principle, we will see scattered light from all along the 'equal light travel time' parabolic surface, but the observed brightness is of course weighted with the scattering probability for the scattering angle implied by a given photon trajectory, and with the density of scattering dust along that surface.

Halo Annular Width

The angular width of a scattering echo can be found as follows. There are several distinct contributions: (1) the length of the observation compared to the time since the event, (2) the width of the angular probability distribution as a function of scattering angle, (3) the telescope angular resolution.

First, consider the length of the exposure compared to the time since the arrival of the unscattered light. Let's now call the time delay, or time since arrival of the unscattered light, t_0 :

$$t_0 = \frac{1}{2} \theta_0^2 \frac{d}{c} \quad (3)$$

so for an exposure length Δt_0 :

$$\frac{\Delta t_0}{t_0} = 2 \frac{\Delta \theta_0}{\theta_0} \quad (4)$$

The range of angles you should see, $\Delta \theta_0$, is related to the length of time you expose, Δt_0 . So let's say you're looking one year after the event: $t_0 = 1$ yr, and you have 1 day worth of data, then $\Delta \theta_0 / \theta_0 = (1/2) \times 1/365$. This basically means you can assume all photons were scattered from the same paraboloid equal-time-delay surface; the size of that paraboloid expands by only 2 parts in 1000 over the timespan of your data. One year after arrival of the unscattered light ($t_0 = 1$ yr), the mean scattering angle is: $t_0 = (1/2) \theta_0^2 (d/c)$; let's assume $d/c = 25000$ years, corresponding to $d = 8.5$ kpc, so we get $\theta_0 \approx \sqrt{8 \times 10^{-5}} = 0.009$ radians = 0.5 deg. And the range of angles implied by the length of an exposure (< 1 day) is negligible: $\Delta \theta_0 = 2.5$ arcsec in this example.

Now the effect of the width of the angular probability distribution for scattering of X-rays. The characteristic angular width of the distribution is approximately (Mauche and Gorenstein, *Ap.J.*, **302**, 371 [1986]): $\sigma = 10.4(a/0.1 \mu\text{m})^{-1}(E/1 \text{ keV})^{-1}$ arcmin, with a the dust grain radius and E the photon energy in keV. The cross section is approximately a Gaussian, $\exp(-\theta^2/2\sigma^2)$. So now consider the scattering at some instant of time after the arrival of the unscattered radiation. Since the scattering

probability drops so steeply as a function of angle, most of the scattered light we'll see will be from locations (where there is dust) that are far away. So once θ is larger than σ , it goes fast.

We'll see some fuzziness implied by the finite width of the scattering probability distribution. Let's say $\sigma = 10.4/2 = 5.2$ arcmin (looking at 2 keV photons, with $a = 0.1$ micron), and $\theta_0 = 0.5$ degrees, where θ_0 is the smallest angle, from the most distant dust. For the probability to drop by factor 10 from the value for θ_0 you need an angle θ defined by:
 $\exp(-\theta^2/2\sigma^2) = 0.1 \exp(-\theta_0^2/2\sigma^2)$, or $\theta^2 - \theta_0^2 = 2\sigma^2/\ln(10)$, so (set $\theta = \theta_0 + \Delta\theta$):

$$\frac{\Delta\theta}{\theta_0} = \ln(10) \frac{\sigma^2}{\theta_0^2} \quad (5)$$

In our case: $\Delta\theta/\theta_0 = 0.064$ or so: one part in 16. That's narrow in relative measure. In absolute measure, it's about 0.06×0.5 degrees = about 2 arcmin. That is much wider than the telescope resolution (about 15 arcsec for eROSITA), so you don't have to worry about that.